EE561 Advanced Static Power Conversion

Multilevel Converters — Quiz Answer Key

For the presenters and the instructor · not for distribution to students

Question 1 · Device selection — 25 pts

(a) Required rating = blocked voltage ÷ (2/3) = 1.5 × blocked voltage; pick the next class.

Leg / device	Blocked voltage	Needed rating	Class
2-level, each switch	1100 V	1650 V	**1700 V**
3L NPC, each switch	550 V	825 V	**1200 V**
3L T-type, outer S1, S4	1100 V	1650 V	**1700 V**
3L T-type, inner S2, S3	550 V	825 V	**1200 V**

(b) Advantage: in P and N only one device conducts (NPC: two) → lower conduction loss; no clamp diodes. Disadvantage: the outer switches must block the full 1100 V → 1700 V class: slower, more switching loss, fewer and costlier parts.

Marking: (a) 3 pts per row + 3 pts for the 2/3 rule; (b) 5 + 5.

Question 2 · NPC leg in the zero state — 25 pts

(a) Path: output a → **S3** → node between S3 and S4 → **D6** → midpoint O. Conducting devices: S3 and D6.

(b) Current flows from the output into O: the leg **pushes current into the midpoint** (current drawn from O is $i_O = i_a < 0$).

(c) S1 blocks 400 V (P to node X, which sits at O through S2) and **S4 blocks 400 V** (node Y at O to N). S2, S3, D5, D6 see ≈ 0 V.

(d) Going P → N directly switches both complementary pairs at once: the output jumps by the full 800 V (twice the dv/dt and EMI of a normal step), and during the transient the series devices S1–S2 or S3–S4 must share 800 V with no clamp guaranteeing equal sharing → one device can be overvoltaged. Nothing is saved: switching loss scales with the voltage step. Always pass through O.

Marking: (a) path 4 + devices 3; (b) 5; (c) 3 + 3; (d) step size 3, voltage sharing 3, conclusion 1.

Question 3 · DC-bus utilization and min-max injection — 25 pts

(a) $V_{1,pk} = 380 \cdot \sqrt{2} / \sqrt{3} = 310.3$ V; $m = 310.3 / 300 =$ **1.034**.

(b) Sine PWM: **no** — $m > 1$, the references exceed the carriers (overmodulation, low-order harmonics). Min-max: **yes** — $1.034 < 2/\sqrt{3} = 1.155$.

(c) max = 0.98, min = −0.73 → $r_0 = -(0.98 - 0.73)/2 =$ **−0.125**; $r' = (0.855, -0.375, -0.855)$. $r'_a - r'_b = 1.23 = r_a - r_b$: the common offset cancels in every line-to-line difference.

(d) $V_{LL,max} = \sqrt{3} \cdot (2/\sqrt{3})(V_{dc}/2) / \sqrt{2} = V_{dc}/\sqrt{2} =$ **424 V** (sine PWM: 367 V).

Marking: (a) 6; (b) 2.5 + 2.5; (c) r_0 3, r' 3, check 3; (d) 5.

Question 4 · Neutral-point balancing **25 pts**

(a) $\langle i_O \rangle = \Sigma(1 - |r_x + r_0|)\, i_x$. For a small offset that does not change the sign of r_x: $|r_x + r_0| = |r_x| + \text{sign}(r_x)\, r_0$, so $\boldsymbol{\Delta\langle i_O \rangle = -r_0\, \Sigma\, \text{sign}(r_x)\, i_x}$.

(b) Averaged over a period, $\Sigma\, \text{sign}(r_x)\, i_x = (6/\pi)\, \hat{I} \cos\phi$. At $\phi = 90°$ this is ≈ 0: the offset no longer changes the average midpoint current, so **ΔV is not restored** (only the $3f_1$ ripple remains). Low power factor needs another method (redundant-vector selection, larger capacitors).

(c) The sensitivity $-(6/\pi)\, \hat{I} \cos\phi$ **changes sign with the power direction**. In rectifier mode ($\cos\phi < 0$) the loop tuned as negative feedback becomes **positive feedback**: ΔV runs away until the overvoltage protection trips. Fix: $r_0 = \text{sign}(P) \cdot K \cdot \Delta V$ (or a current-direction-aware state selection), and test both power directions.

Marking: (a) 8; (b) prediction 3, explanation via $\cos\phi$ 5; (c) sign change 4, consequence 2, fix 3.